

Hyper Efficient Theremin ,a Novel CD4093BE + MCP602 Design

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Abstract— This project is a theremin built using a CD4093BE NAND Schmitt trigger IC and an MCP602 dual op-amp. Two NAND gate oscillators generate frequencies that are mixed and filtered to produce an audible tone. Hand proximity to a copper wire antenna shifts the capacitance, changing the beat frequency heard through the speaker. A second op-amp amplifies the signal and drives an 8 Ω speaker on a single 9V supply. The design avoids inductors entirely, showing how standard digital logic ICs can be used for analog audio synthesis in a simple, low cost circuit.

Keywords— CD4093BE, MCP602, NAND Schmitt trigger, beat frequency, capacitive sensing, op-amp, low-pass filter, audio synthesis, hand proximity, 9V supply

I. INTRODUCTION

This project explores the design and implementation of a contactless musical instrument using the CD4093BE quad NAND Schmitt trigger IC and the MCP602 dual op-amp. The theremin is one of the few instruments played without physical contact, relying on the capacitive effect of the human hand near an antenna to control pitch. In this design, two NAND gate oscillators generate high frequency signals whose difference falls in the audible range. This beat frequency varies as the hand moves closer or farther from the antenna. The mixed signal is then filtered and amplified to drive a small speaker. The circuit operates on a 9V supply and does not use any inductors, making it a compact and accessible build. This report covers the working principle, component selection, circuit stages, and observed results of the project.

II. COMPONENTS AND CIRCUIT DESIGN

The CD4093BE is a quad two input NAND Schmitt trigger IC. The Schmitt trigger inputs give it hysteresis, which makes it reliable for building RC oscillators even with slow rising edges. Two of its four gates are configured as oscillators, one variable and one fixed reference. A third gate acts as a mixer and the fourth buffers the output.

The MCP602 is a dual op-amp with rail to rail output, well suited for single supply operation at 9V. The first op-amp is configured as an inverting integrator to form a low pass filter that removes the high frequency RF components and passes only the audio envelope. The second op-amp amplifies the filtered signal with a gain of 11 to drive the speaker.

Together these two ICs handle oscillation, mixing, filtering, and amplification, covering the full signal chain with minimal components.

III. WORKING PRINCIPLE AND DESIGN CALCULATIONS

A.Principle

The theremin works on the principle of beat frequency detection. When two oscillators running at slightly different frequencies are mixed, they produce a third signal whose frequency is the difference of the two. In this circuit, Gate 1 of the CD4093BE acts as a variable oscillator running at around 833 kHz, and Gate 2 acts as a fixed reference oscillator at around 122 kHz.

When a hand approaches the copper wire antenna, it adds a small capacitance of roughly 1 to 10 pF to the Gate 1 RC network. This shifts Gate 1's frequency slightly. Gate 3 mixes both oscillator outputs, and the difference between them falls in the audible range of 200 Hz to 5 kHz depending on hand distance.

The mixed signal still contains high frequency RF components, so it passes through the MCP602 op-amp configured as a low pass filter with a cutoff of around 16 Hz, which extracts only the slowly varying audio envelope. A second op-amp stage amplifies this with a gain of 11 and drives the speaker through a 4.7 microfarad coupling capacitor that blocks any DC offset.

The result is a pitch that rises as the hand moves closer to the antenna, with no physical contact needed at any point.

B. Calculations

The circuit design involves three main calculations.

Oscillator frequency-

Using $f = 1/(1.2 \times R \times C)$, Gate 1 with 10k ohm and 100pF gives a base frequency of approximately 833 kHz. Gate 2 with 6.8k ohm and 1nF gives a fixed reference of approximately 122 kHz. The beat frequency is the difference between the two, which varies as hand proximity changes Gate 1's frequency.

Frequency Sensitivity-

The shift in frequency per picofarad of hand capacitance is given by $\Delta f = f \times (\Delta C / C)$. At 833 kHz with $C = 100\text{pF}$, this gives roughly 8 kHz per picofarad. Since hand capacitance ranges from 1 to 10 pF, the total frequency shift can reach up to 80 kHz, which comfortably covers the audible beat range.

Low Pass Filter-

The cutoff frequency of the integrating low pass filter is $f = 1/(2 \times \pi \times R \times C)$. With $R = 100\text{k ohm}$ and $C = 100\text{nF}$, the cutoff is approximately 16 Hz. This removes all RF components and passes only the audio envelope.

Amplifier Gain-

Op-amp B is configured as a non inverting amplifier with gain $G = 1 + (R_f/R_{in}) = 1 + (100\text{k}/10\text{k}) = 11$. With a voltage swing of around 4V, the output power into the 8 ohm speaker is approximately 50 mW.

IV. RESULT AND OBSERVATIONS

The circuit was assembled on a breadboard and powered using a 9V battery. Upon switching on, a steady tone was heard from the speaker confirming that both oscillators were running and the beat frequency was falling in the audible range.

When the hand was brought closer to the antenna, the pitch increased noticeably, and moving the hand away lowered it back. The response was smooth and continuous, with no sudden jumps in frequency. The usable range of hand movement was roughly 5 to 25 cm from the antenna tip.

The potentiometer successfully controlled the output volume without affecting pitch. The 4.7 microfarad coupling capacitor eliminated the DC offset at the speaker, which improved sound clarity noticeably compared to a direct connection.

Some instability was observed when the 9V battery leads were kept long, causing slight frequency drift. Shortening the power connections resolved this. The circuit was also sensitive to nearby metal objects, which slightly shifted the base pitch.

Overall the design performed as expected, producing a contactless pitch controlled output using only two ICs and passive components on a single 9V supply.

A. Figures and Tables

a) *Positioning Figures and Tables*: Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation "Fig. 1", even at the beginning of a sentence.

TABLE I. TABLE TYPE STYLES

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^a Sample of a Table footnote. (Table footnote)

Fig. 1. Example of a figure caption. (figure caption)

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